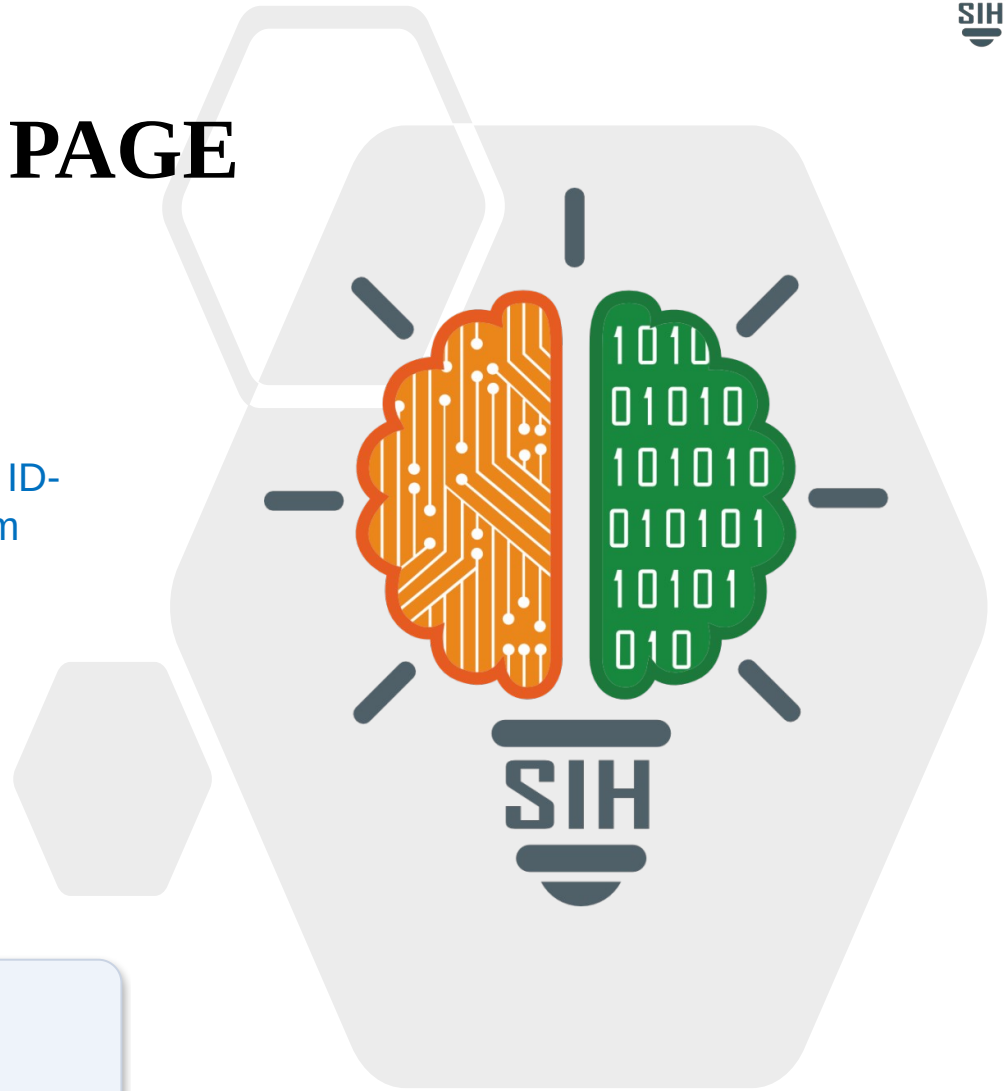
SMART INDIA HACKATHON 2026

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TITLE PAGE



- **Problem Statement ID –** SIH26263
- **Problem Statement Title-** Development of an APAAR ID-Based Student Data Management and Verification Platform for AICTE-Approved Institutions
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-**
- **Team Name (Registered on portal)** AI Riders

VIDYA PRAMAAN

One student. One APAAR ID. One verifiable record.

Working prototype: https://vidya-pramaan-sih26263.onrender.com

AI Riders

VIDYA PRAMAAN

29.6 lakh enrolments in AICTE institutions (2025-26) · **24.25 crore** APAAR IDs issued (Mar 2026)

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❖ Proposed Solution (Describe your Idea/Solution/Prototype)

❖ Detailed explanation of the proposed solution



Institution upload
Excel · CSV · ERP API

APAAR validation
registry + consent

Data verification
31 explained rules

Central repository
one record per student

Analytics & monitoring
live, case by case

Authorised verification
consent + signed QR

❖ How it addresses the problem

✖ Only total enrolment reaches AICTE (except PGDM), per the PS	✔ Student-level records keyed on APAAR, live
✖ Excess admission: fine of 5× the fee per student (APH)	✔ Seat #61 of 60 blocked at upload and at submit
✖ Same student enrolled at two colleges goes unseen	✔ Cross-college match + shared-contact rings
✖ Records are hard to verify at source (PS)	✔ Signed QR (checks offline) + consent for more

❖ Innovation and uniqueness of the solution

Twin-safe matching
Mohd = Mohammed, but Rohit ≠ Mohit

Ghost-admission map
1 phone, 8 students, 2 colleges: flagged

Decisions kept
reason required; never re-flagged

Signed checkpoints
even a full history rewrite is caught

Rules as data
2026-27 caps by level, each with its source

APAAR on degrees
university print feed with verification QR

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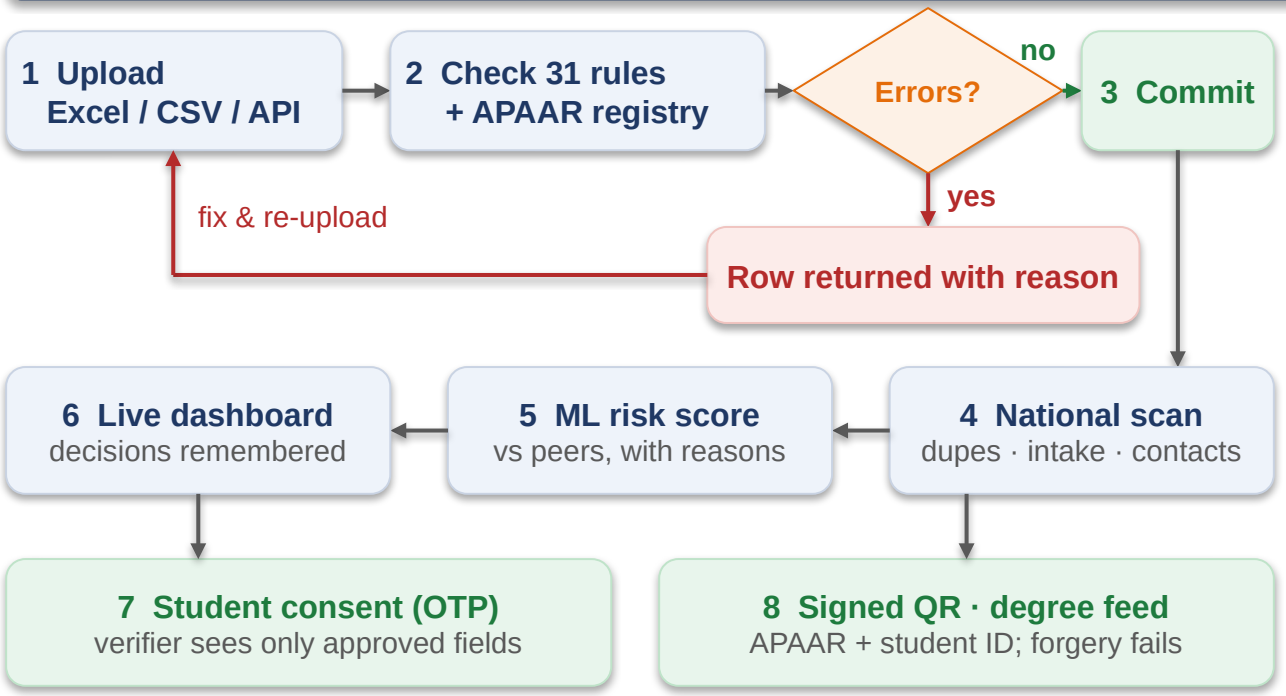
TECHNICAL APPROACH

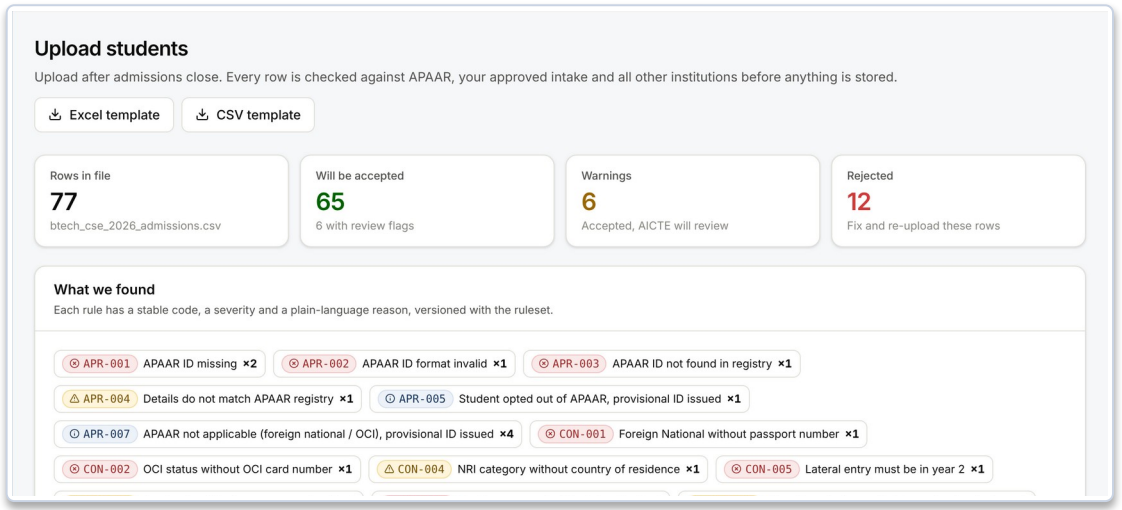
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❖ Technologies to be used (e.g. programming languages, frameworks, hardware)

- **Python · FastAPI** — REST · OpenAPI
- **React · TypeScript** — installable PWA
- **PostgreSQL 16** — SQLite in the demo
- **scikit-learn** — Isolation Forest
- **Name matcher** — twin-safe · Indic
- **Ed25519 · W3C** — signed QR + audit

❖ Methodology and process for implementation (Flow Charts/Images/ working prototype)





Working prototype: 77-row upload → 65 accepted, 12 returned with rule codes

Try it live: https://vidya-pramaan-sih26263.onrender.com
7 demo roles · synthetic data · installable on phones

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FEASIBILITY AND VIABILITY

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Analysis of the feasibility of the idea

0.79 / 0.92
held-out precision / recall (old matcher 0.65 / 0.68)

0 / 80
same-day twins wrongly merged

17/17
planted error types; 0 false flags on 56 clean rows

164 · 57 · 4
planted dual enrolments, identity dupes, contact rings

42/46
top-ranked colleges truly anomalous

59 tests
SQLite + PostgreSQL 16; 7-role browser walkthrough

Planted cases are regression checks on synthetic data, not real-world accuracy · precision / recall come from independent hand-labelled name pairs

Potential challenges and risks	Strategies for overcoming these challenges
No public APAAR API	✔ Registry adapter: mock now → DigiLocker / API Setu after partner onboarding
Students without APAAR (opt-out, FN / OCI)	✔ Provisional ID, never rejected; mother's or guardian's name accepted
Student privacy (DPDP Act; APAAR is voluntary)	✔ Consent per field, masked IDs, no Aadhaar stored, access log
Wrong flags: twins, spelling variants	✔ Twin-safe matcher; officer decides with a reason; decision remembered
Rules change every year by circular	✔ Rules file per academic year, each value with its source
ERP formats, low digital readiness	✔ ERP headers auto-mapped, Excel template, API keys, Hindi, phone app

Roadmap M1–3 pilot: 1 state, 50 colleges → M4–6 API Setu, SSO, STQC audit → M7–12 national **Cost** open-source stack, no licence fees; the whole prototype runs on one free 512 MB instance

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IMPACT AND BENEFITS

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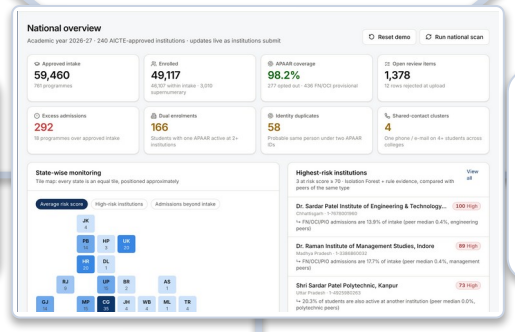
❖ Potential impact on the target audience

AICTE & regulators
excess & dual enrolment caught at submission

Institutions
one upload; errors and corrections explained

Universities
APAAR + student ID printed on degrees

Employers
instant QR check, even offline; consent for more



One national view

Students
own record; control over who sees it

Today
✖ 42 excess seats found late: ₹23.1 cr fine (FORE v AICTE, SC 2019)

✖ AISHE 2022-23 report published only in 2026

✖ "a number of students have mismatch in Aadhaar and APAAR ID details" (CU Srinagar, 2026)

With Vidya Pramaan
✔ Seat #61 of 60 stopped the day it is uploaded, before any fine

✔ Same-day national view; duplicates flagged within minutes

✔ Mismatches returned with the reason, fixed by re-upload

❖ Benefits of the solution (social, economic, environmental, etc.)

Social
APAAR opt-outs never refused, as the Supreme Court requires

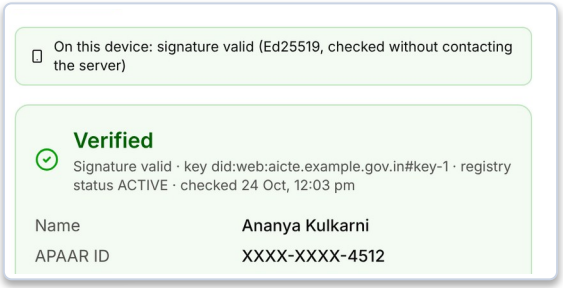
Economic
MIT / BSD / Apache stack: no licence fees; one upload, all users

Environmental
paper returns and verification letters replaced by upload + QR

Governance
audited evidence pack for review and show-cause



Certificate: APAAR ID + student ID + QR



QR scan: verified, also on the device

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RESEARCH AND REFERENCES

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❖ Details / Links of the reference and research work

Why not just use the systems that exist?

	NAD / DigiLocker	ABC	AISHE	Vidya Pramaan
Student-level admissions vs approved intake	—	—	aggregate, 2–3 yr lag	✔ live
Same student at two colleges; ghost admissions	—	—	—	✔
Supernumerary caps by programme and year	—	—	—	✔
Consent-based verification for employers	✔	—	—	✔ field-level
APAAR + student ID feed for degree printing	issuer uploads	credits only	—	✔

Vidya Pramaan consumes APAAR and feeds NAD / ABC and AISHE: no new ID, no parallel credential store, no invented APAAR checksum (none is published), and no student rejected for opting out.

References

1. AICTE PS SIH26263, SIH 2026 sih.gov.in
2. AICTE Approval Process Handbook 2024-27 and 2026-27 addendum (FN/OCI/PIO caps) aicte.gov.in
3. Supreme Court, FORE School of Management v AICTE (2019): 5× fee per excess student sci.gov.in
4. NAD on DigiLocker: consent-based verification (FAQ) nad.digilocker.gov.in
5. AISHE 2022-23 & 2023-24 reports (released 2026) aishe.gov.in
6. APAAR opt-out: Orissa HC, Rohit Anand Das (Dec 2025); Supreme Court direction (Jul 2026) scconline.com · livelaw.in
7. Parliament replies: AICTE enrolment 2025-26 (Lok Sabha); APAAR IDs at 30 Mar 2026 (Rajya Sabha) sansad.in
8. Digital Personal Data Protection Act 2023; DPDP Rules 2025 meity.gov.in
9. W3C Verifiable Credentials; RFC 8032 (Ed25519) w3.org · rfc-editor.org
10. Liu, Ting & Zhou (2008) Isolation Forest; IDinsight on matching Indian names doi.org/10.1109/ICDM.2008.17 · idinsight.org

Live prototype
https://vidya-pramaan-sih26263.onrender.com
Synthetic data only · 7 demo roles · works on phones

Source code & tests
https://github.com/Pulkitkh/VidyaPramaan
59 automated tests + browser walkthrough · MIT licence